



THINGS YOU WANTED TO KNOW!

Ques. I TRIED USING POWERBANKS FROM BOTH ASUS AND XIAOMI TO POWER AN ELECTRONIC DIYA (LAMP) THAT HAS THREE LEDs. WHEN I CONNECTED THE POWERBANK, LAMP LEDs GLOWED AND POWERBANK LED INDICATOR STARTED BLINKING. HOWEVER, LAMP LEDs WENT OUT AFTER A SHORT INTERVAL. WHAT COULD BE THE PROBLEM?

Bala Giridhar

Ans. Forward voltage of each white LED varies between 3V and 5V. A powerbank normally gives 5V DC output. Depending on the type of series or parallel connections, an electronic *diya* with three white LEDs would require a 7.5V to 12V DC source. Try using a powerbank that has an output capacity of 12V DC.

Q2. HOW CAN I DESIGN AUDIO POWER TRANSFORMERS? WHERE CAN I GET THESE?

Jogindar Kumar

A2. Audio power transformers are used in sound amplifiers, guitar amplifiers, super reverbs, public address systems and so on. These are of different types depending on applications like audio power supply, impedance matching and isolation. Depending on types, steps for designing these vary.

Power audio transformers operate at low frequencies, such as 50Hz or 60Hz. Although, some operate over the audio frequency range of 20Hz to 20kHz. Covering these topics is not possible here, so we have narrowed it down to low-frequency-range transformers used in power supplies.

A power supply is one of the most important parts of an audio system. A step-down transformer is mostly used in an audio power supply. Here, primary winding is driven with AC line voltage. Voltages required for the rest of the audio circuit are generated in secondary windings of the transformer.

Almost all power transformers share some characteristics. Windings are placed on a core made of iron alloy. To design a

transformer that is nearly 100 per cent efficient, ensure that all energy in primary winding is coupled to the secondary, and is not wasted by heating up the core or windings, or leaking magnetic flux outside the core. Also, make it inexpensive to build. Very often the most critical part is the amount of hum that the transformer induces into the electronic circuitry.

For designing a power transformer for audio equipment, follow the steps given below.

1. Note your requirements.
2. Select a core size.
3. Compute primary turns.
4. Compute secondary turns.
5. Fit wire to core window.
6. Get required components.
7. Wind primary and secondary coils.
8. Stack laminations.
9. Test performance.
10. If result is ok, good; otherwise, check the parameters and start again with suitable core size.

Requirement. Include input and output voltage and current. Input requirement is 230V AC in primary winding. Then, decide output current and voltage in secondary winding.

Selecting the core. While EI-core (its laminated core looks like the letters E and I) is the least expensive and most common type of transformer, it is also one of the worst in terms of its stray flux. This is because there is air gap between laminations in E to I interfaces. If cost is not the deciding factor, a toroidal transformer provides superior performance as compared to EI-core.

Windings. Determine primary winding current and wire size. Primary current should be equal to total output power plus transformer power losses, divided by primary voltage. Next, determine the number of turns required for secondary winding.

Verification. Verify the design by measuring the open-circuit and loaded voltages across secondary winding, by calculating resistance of each winding. Then, calculate voltage drop across that winding by multiplying resistance and current in the winding. Now, calculate

open-circuit and loaded voltages across secondary winding, using the formulae given on the websites mentioned below.

Calculating temperature. Acceptable temperature rise depends on the application and the designer. There are two main causes of temperature rise in a transformer: core power loss and winding power loss. These losses can be calculated using standard formulae. You may check further details from the following websites:

www.electronicdesign.com

www.ti.com

www.audioxpress.com

www.eetimes.com

Power transformers are available online at www.ebay.com, www.element14.com, www.mouser.in and other stores.

Q3. HOW CAN I AUTO-ROUTE PCB LAYOUT TRACKS IN GEDA SOFTWARE ON A 45-DEGREE ANGLE?

A. Samiuddhin

A3. There is no automatic way of making the track bend at 45 degrees in auto-route mode. You can achieve it manually by following the steps given below.

1. Open gEDA PCB Designer.
2. Import schematics of any circuit made on gEDA Schematic Editor.
3. On top of the screen, go to Select menu and choose Disperse all elements.
4. Optimise rats nest to connect all components used in the PCB, as per the circuit diagram. For this, press o on the keyboard, or choose Connects and click on Optimise rats nest. Place the components by dragging and dropping (using a mouse), as per requirement.
5. Again, go to Connects and click on Auto-route all rats. All tracks will get connected to the components automatically.
6. Bending of tracks is set at 90 degrees by default. For the track to bend at 45 degrees instead, select the required layout tracks. Select Optimise Routed Tracks and click on Miter.

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